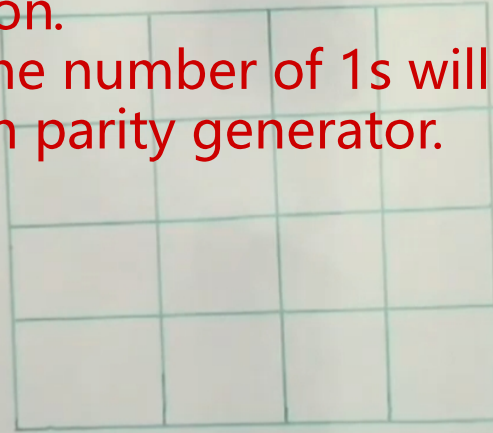


4-Bit Even Parity Generator

B_3	B_2	B_1	B_0	P
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

*Why Even Parity when it works upon odd number of 1s ?
- It focuses on the final result, instead of intermediate action.

- The number of 1s will always be Even in result in case of Even parity generator.



=> Generates parity **1** for
odd number of **1s** in input

4-Bit Even Parity Generator

B_3	B_2	B_1	B_0	P
0	0	0	0	0 (0)
0	0	0	1	1 (1)
0	0	1	0	1 (2)
0	0	1	1	0 (3)
0	1	0	0	1 (4)
0	1	0	1	0 (5)
0	1	1	0	0 (6)
0	1	1	1	1 (7)
1	0	0	0	1 (8)
1	0	0	1	0 (9)
1	0	1	0	0 (10)
1	0	1	1	1 (11)
1	1	0	0	0 (12)
1	1	0	1	1 (13)
1	1	1	0	1 (14)
1	1	1	1	0 (15)

$B_3 B_2$

$B_1 B_0$	00	01	11	10
00		1		1
01	1		1	
11		1		1
10	1		1	

✓ Check-Board Configuration

$$P = B_0 \oplus B_1 \oplus B_2 \oplus B_3$$

$$P = \overline{B_3} \overline{B_2} \overline{B_1} B_0 + \overline{B_3} \overline{B_2} B_1 \overline{B_0} + \overline{B_3} B_2 \overline{B_1} \overline{B_0} + \overline{B_3} B_2 B_1 B_0 + B_3 \overline{B_2} \overline{B_1} \overline{B_0} + B_3 \overline{B_2} B_1 B_0 + B_3 B_2 \overline{B_1} \overline{B_0} + B_3 B_2 B_1 B_0$$

$$P = \overline{B_3} \overline{B_2} (\overline{B_1} B_0 + B_1 \overline{B_0}) + \overline{B_3} B_2 (\overline{B_1} \overline{B_0} + B_1 B_0) + B_3 \overline{B_2} (\overline{B_1} B_0 + B_1 \overline{B_0}) + B_3 B_2 (\overline{B_1} \overline{B_0} + B_1 B_0)$$

$$= [\overline{B_3} \overline{B_2} (B_0 \oplus B_1) + \overline{B_3} B_2 (\overline{B_0} \oplus \overline{B_1}) + B_3 \overline{B_2} (B_0 \oplus B_1) + B_3 B_2 (\overline{B_0} \oplus \overline{B_1})]$$

$$= (B_0 \oplus B_1) (\overline{B_3} \overline{B_2} + B_3 B_2) + (\overline{B_0} \oplus \overline{B_1}) (\overline{B_3} B_2 + B_3 \overline{B_2})$$

$$= (B_0 \oplus B_1) (\overline{B_2} \oplus B_2) + (\overline{B_0} \oplus \overline{B_1}) (B_2 \oplus B_2)$$

$$= A \cdot \overline{B} + \overline{A} \cdot B$$

$$P = B_0 \oplus B_1 \oplus B_2 \oplus B_3$$